

# PSU\_USB - v1.0 - SSS SSDD - SC25

Project: PSU\_USB

Version: 1.0

Object: System and Subsystem Specification, System And Subsystem Design Description

Design Season: SC25

## Document Overview

This document describes the specifics design decisions for the PSU\_USB board, the object of this document is to describe functional and non-functional requirements. Each requirement must be a single testable unit.

## Acronym and Abbreviations

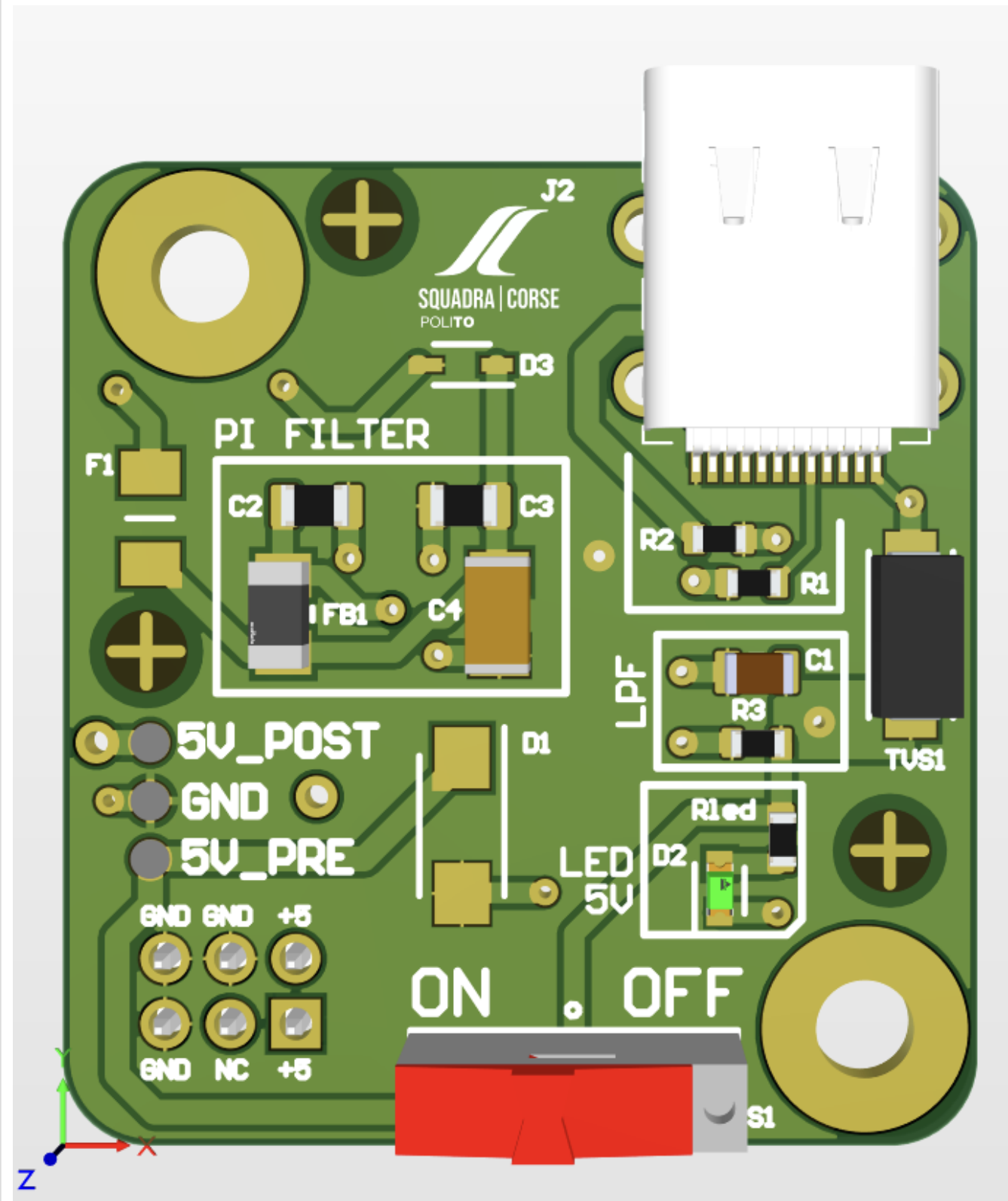
| Symbol | Meaning                       |
|--------|-------------------------------|
| PSU    | Power Supply                  |
| USB    | Universal Serial Bus          |
| PD     | Power Delivery                |
| IC     | Integrated Circuit            |
| ESD    | Electro Static Discharge      |
| MLCC   | Multi Layer Ceramic Capacitor |

## System Overview

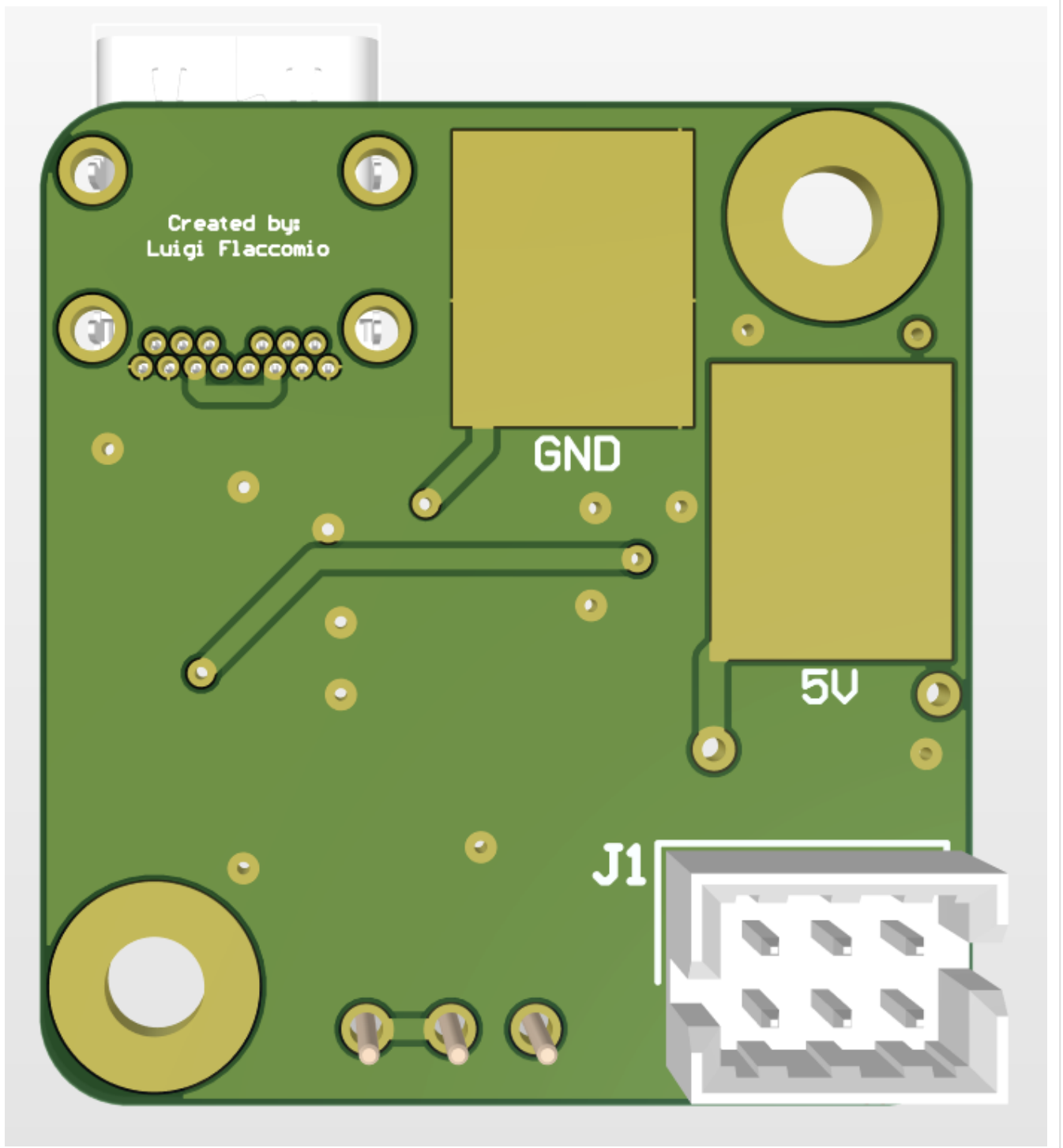
The PSU\_USB board is meant as a useful device to power up boards that requires stable 5V power supply. Using a USB-C connector is capable of powering up electronics board with ease, connecting to any power supply or a PC.

Board 3D View

Top View



Bottom View



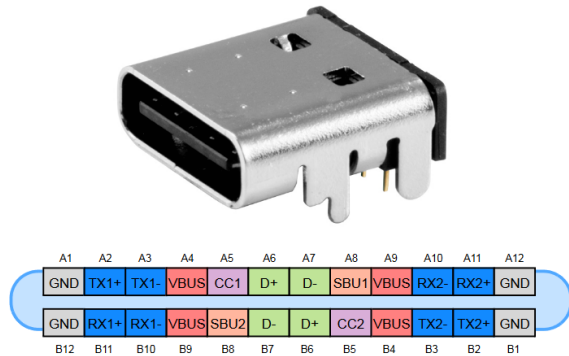
## Referenced Documents

| Reference                    | Name                   | Source                     | Description                              |
|------------------------------|------------------------|----------------------------|------------------------------------------|
| <a href="#">632723300011</a> | USB-C Female Connector | <a href="#">Data Sheet</a> | Data Sheet of the USB-C female connector |
| SMAJ30CA-TR                  | TVS Diode              | <a href="#">Data Sheet</a> | Data Sheet of the TVS Diode              |

PSU\_USB board can be connected to any power source via USB-C connector. The cable will then only supply 5V to the circuit due to USB standard limitation.

## USB-C connection

USB-C is a well known standard for data transferring and power application. A very important aspect of this connector is the possibility to use a Power Delivery mode that increases the power that can be delivered to the device. A pinout of the connector can be seen in the picture below:



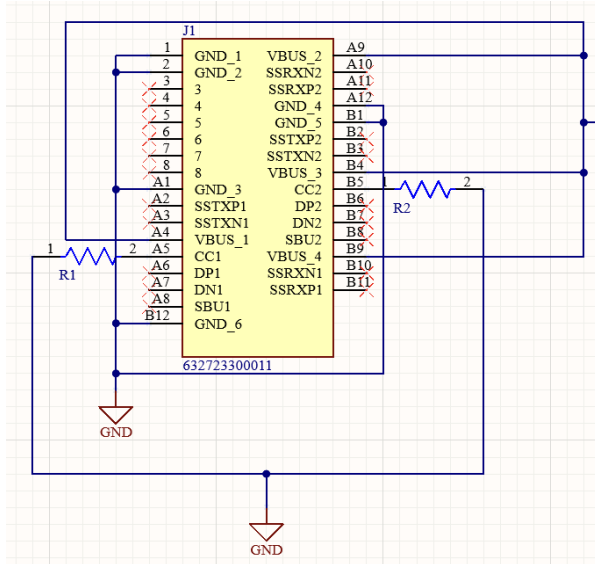
Normal USB devices, without the use of a negotiation IC can only supply 5V at 3A, making up a power of only 15W. With the use of a PD Negotiation Chip, instead, is possible to increase this value by a lot, making it possible to power up a device with power that can easily arrive to 240W with the latest standard USB 3.1.

In this specific application, since only a 5V was required, it has been decided to not use a PD negotiation chip. By doing this, the complexity of the entire system was simplified and board space was optimized.

The only constraint that is required to not use a PD IC is to use 2 Pull Down Resistors connected to the CC1 and CC2 pins of the connector. By doing this, the connector is limited to 5V 3A (15 W) of maximum output. Also, for better clarity, a list of all the connections can be done:

- **VBUS pins:** For power applications like in this case, all VBUS pins can be connected together and contribute to the power output;
- **CC1 and CC2 pins:** To limit the voltage to 5V, two 5.1 kOhms resistors have been connected from each pin to GND reference;
- **GND pins:** All ground pins are connected together to a ground reference;

- All other pins are left unconnected for this application;



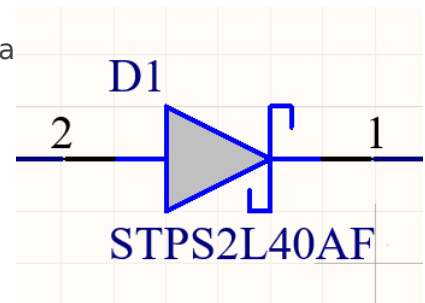
## Internal Protection Circuits

PSU\_USB has different protection circuits to avoid problems during working hours. All the circuits are described in sections below.

### Protection Circuits

#### Reverse current protection

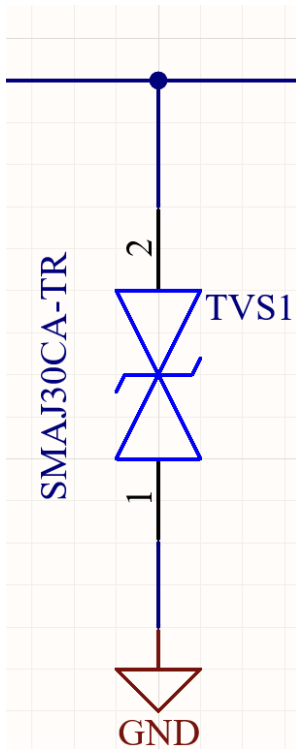
A Zener diode with low voltage drop is used to protect the circuit a



can occur when the circuit is connected to a device via USB cable.

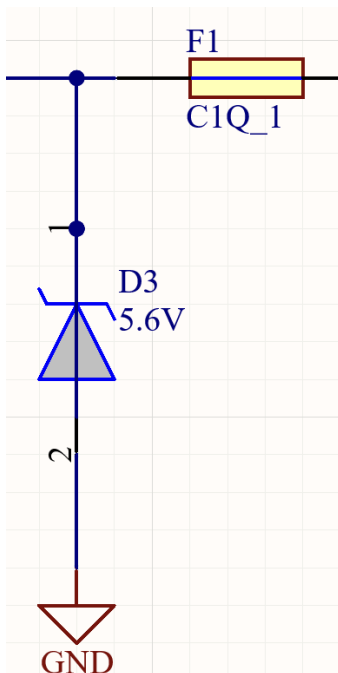
#### ESD Protection

The circuit must be protected against ESD. This can be achieved by using a TVS diode that have fast response and can repel voltages peaks from both directions.



## Overvoltage and Overcurrent Protection

The most important protection circuit is against Overvoltage and overcurrent that can occur during usage. In this case a diode and a fast blow fuse do the job.



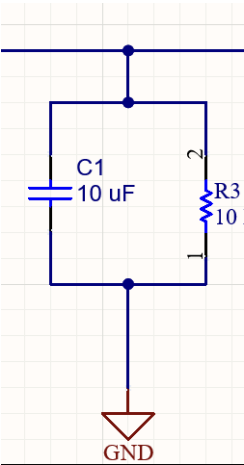
## Filtering Circuits

### Filters

To guarantee a stable 5V supply, filters must be used. In this case 2 filters are used in the circuit.

### LPF Filter

In order to avoid bouncing due to the switch S1 used in the circuit, a LPF filter is used. In this case the filter also helps with improving the stability of the final output of the circuit, working together with the other PI filter. With the components used (C1: 10uF and R3: 10kOhm) a 1.6Hz cutoff frequency can be realized, with a time constant of about 100 ms.



### PI Filter

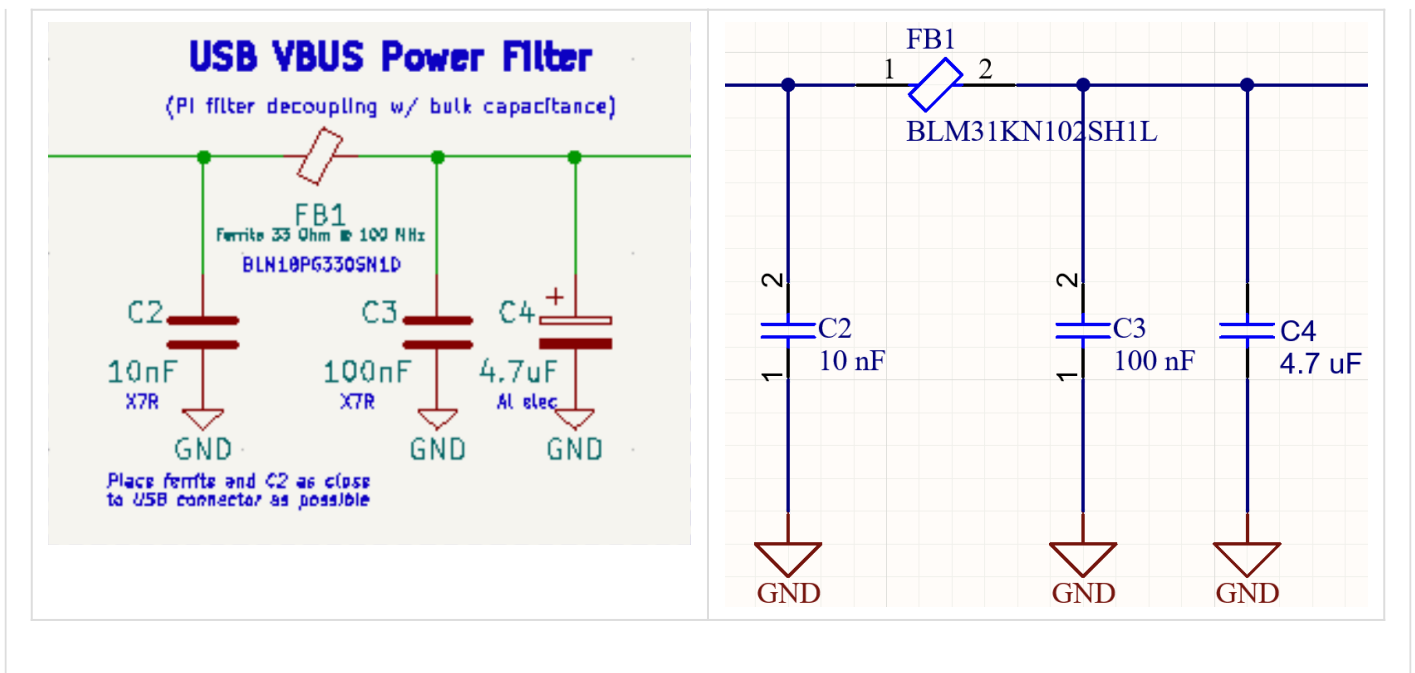
The PI Filter is a very common configuration used in power application to guarantee a stable output. In this case, since the USB-C connector could be connected to any power source (PC or other power supply), a very important role is played by this filter in avoiding noise in the output voltage. The values chosen in this case have been taken from an existing configuration specifically used in USB application, with some differences:

- C4, in the original configuration, should be an Electrolytic capacitor with low ESR. This because it serves as a bulk where the noise can go to avoid disturbances in the circuit. After some considerations, since an electrolytic capacitor would have been too big, a MLCC with low ESR and much less impact in terms of occupied space has been chosen. This capacitor makes the circuit work worse, but the trade offs between space occupation and performance have been better satisfied.
- C2 and C3 have been left unchanged from the original configuration since the working principle was fulfilled.

A comparison between the original configuration and the one chosen can be observed below:

| Original Configuration | Chosen Configuration |
|------------------------|----------------------|
|------------------------|----------------------|





# Testing Procedures

In order to correctly test the functions of the PSU\_USB board, some steps can be followed. It is important to notice that 3 different testing points (**GND**, **5V PRE**, **5V POST**) are present on the top layer of the board, allowing for easier testing procedures.

## Testing instruments

To test the board a **multimeter** is needed. In this case, the multimeter has to be **set in voltage measurement configuration**.

## Testing steps

The testing procedure has to be performed like it follows:

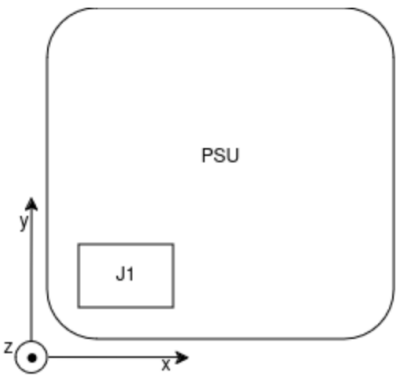
- The **negative multimeter lead** has to be maintained fixed to the **GND testing point** on the board;
- The **positive multimeter lead** has to be moved between the 2 remaining points (**5V PRE** & **5V POST**);
  - To test the correct 5V output of the USB-C connector, the positive lead has to be put onto the 5V PRE testing point;
  - To test the correct functioning of the filters and protection circuit, the positive lead has to be put onto the 5V POST testing point;

## Expected results

In order for the board to work properly, the result of each measurement done on the **5V PRE** and the **5V POST** testing points has to be the same: 5 Volts of stable DC Voltage should be read on the multimeter screen.

# System External Interface Requirements

## Interface Identification and Diagrams



### J1 - Connector

A schematic diagram of the J1 connector. It is a 2x3 pin connector. Pin 1 is connected to +5V. Pin 2 is connected to +5V. Pin 3 is connected to GND. Pin 4 is connected to +5V. Pin 5 is connected to GND. Pin 6 is connected to GND. The connector is labeled J1 and has the part number 87831-0620.

A photograph of the J1 connector pins. The pins are arranged in a 2x3 grid. The pins are labeled as follows: 6 GND, 4 GND, 2 +5, 5 GND, 3, 1 +5. The pin labeled 1 +5 is highlighted with a red border.

Device Side (J1):  
Molex 2x3 straight milligrid Male Connector, PN: 87831-0620

Connector that attaches to PSU on the PCB:  
Molex 2x3 straight milligrid Female Connector, P/N: 79107-7002

| PIN # | Signal Name | Function                         | Type |
|-------|-------------|----------------------------------|------|
| 1     | +5          | 5 Volts from the USB-C Connector | OUT  |

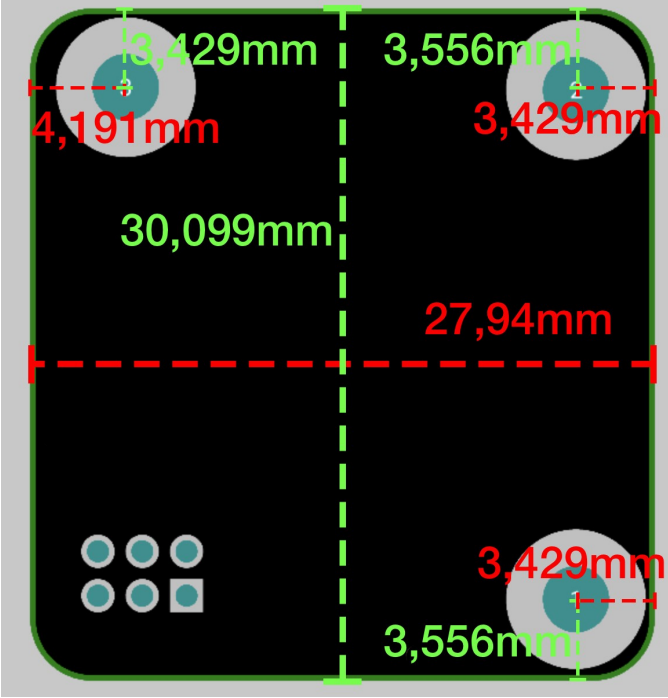
| PIN # | Signal Name | Function                         | Type   |
|-------|-------------|----------------------------------|--------|
| 2     | +5          | 5 Volts from the USB-C Connector | OUT    |
| 3     | /           | Not Used                         | /      |
| 4     | GND         | Ground Reference                 | IN/OUT |
| 5     | GND         | Ground Reference                 | IN/OUT |
| 6     | GND         | Ground Reference                 | IN/OUT |

# System Environment Requirements

The temperature of the PSU\_USB board has to be in the range between -40°C to 85°C.

# Design And Construction Constraints

## Dimension Requirements



The diagram shows a black rectangular board with dimensions 27.94mm (width) and 30.099mm (height). It features three circular holes: Hole 1 at (24.511mm, 3.556mm), Hole 2 at (24.511mm, 26.543mm), and Hole 3 at (4.191mm, 26.670mm). A USB-C connector is located at (6.934mm, 3.715mm). The board also has a 2x3 grid of soldering pins at the bottom left and a square component at the bottom right. Dimensions are labeled in green (3.429mm, 3.556mm, 30.099mm, 27.94mm) and red (4.191mm, 3.429mm, 3.429mm, 3.556mm).

### Dimensions

- Horizontal (x): 27.94mm
- Vertical (y): 30.10mm
- Depth (z): TODO

### Coordinates

- Hole 1 - X: 24.511mm; Y: 3.556mm
- Hole 2 - X: 24.511mm; Y: 26.543mm
- Hole 3 - X: 4.191mm; Y: 26.670mm
- Connector - X: 6.934mm; Y: 3.715mm  
Positioned with the soldering Pins on theComponent side, and the actual connector on the back.